

NETWORKS AND COMPLEXITY

Solution 16-10

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 16.10: Subcritical transcritical bifurcation [Lvl. 4]

We don't usually think of the transcritical bifurcation as having a subcritical and supercritical form. However, in Chap. 15 the transcritical bifurcation behaved in a subcritical way, causing a discontinuous phase transition. Consider the transcritical normal $\dot{x} = ax + bx^2$. Assume that the model demands $x \geq 0$ and find the conditions under which the bifurcation behaves subcritically.

Solution

Close to the bifurcation point the system should behave like the transcritical normal form

$$\dot{x} = ax + bx^2 \quad (1)$$

where the bifurcation point is now at $a = 0$ the steady state at $x^* = 0$ is stable before the bifurcation ($a < 0$). For the bifurcation to show supercritical behavior we need another steady state to be present at $a > 0$. In other words we want a stable, positive steady state to exist for $a > 0$.

In our analysis of the normal form we saw that the second steady state is

$$x_2^* = -\frac{a}{b} \quad (2)$$

and it is always stable if $a > 0$. However the steady state is only positive for $a > 0$ if $b < 0$. To phrase this in terms of the original system we recall that

$$b = \frac{f_{xx}}{2}. \quad (3)$$

Hence the bifurcation behaves supercritically if $f_{xx} < 0$ and subcritically if $f_{xx} > 0$.

To test this result we can once again consider the SIS-model,

$$\dot{x} = px(1 - x) - x \quad (4)$$

in which we observed a supercritical transcritical bifurcation at $x = 0$ and $p = 1$. We compute

$$f_{xx} = \partial_x \partial_x px(1 - x) - x|_* = -2 \quad (5)$$

which is consistent with the supercritical behavior of the transcritical bifurcation.